Special instructions for students

Unless otherwise stated the following parameters have the following definitions:

V_{DS} *: Drain Source Voltage.*

V_{GS} *: Gate Source Voltage.*

V_{TH} *: Threshold Voltage.*

U_t *: Thermal Voltage.*

n *: Weak Inversion Slope factor.*

g_m*: Transconductance.*

Solutions

1. a) Discuss the benefits of both analogue and digital signal processing with regards to power consumption, computational capability and resolution.

[bookwork]
- Analogue - Power consumption depends on supply and biasing current [1]
- Analogue - Computational capabilities arise from the physics of devices [1]
- Digital - Power consumption is dynamic and dependent on the clock [1]
- Digital - Computation arises from well described Boolean logic [1]
- ASP is beneficial when resolution requirements are less that 8 bits due to SNR.[1]
- DSP is beneficial for higher resolution systems when area and noise become more optimum[1],

[6]

b) The differential pair transconductor shown in Figure 1.1 has MOS transistors which are operating in the weak-inversion region of operation with a slope factor n = 1.23.

i) Show that the output current $I_{out} = I_1 - I_2$ of the differential pair transconductor may be given by equation 1 and derive the transconductance efficiency. Show all your steps in derivation.

$$I_{out} = I_{tail} \tanh\left(\frac{V_{GS1} - V_{GS2}}{2nU_t}\right) \quad (1)$$

[4]

Mathematical derivation:

$$I_1 = I_0 \exp\left(\frac{V_{GS1}}{nU_t}\right)$$

$$I_2 = I_0 \exp\left(\frac{V_{GS2}}{nU_t}\right)$$

$$I_{tail} = I_1 + I_2 = I_0(e^{V_{GS1}/nU_t} + e^{V_{GS2}/nU_t})$$

$$I_{out} = I_1 - I_2 = I_0(e^{V_{GS1}/nU_t} - e^{V_{GS2}/nU_t})$$

$$V_{inD} = V_{GS1} - V_{GS2}$$

$$\frac{I_{out}}{I_{tail}} = \frac{e^{V_{GS1}/nU_t} - e^{V_{GS2}/nU_t}}{e^{V_{GS1}/nU_t} + e^{V_{GS2}/nU_t}} = \frac{e^{V_{inD}/nU_t} - 1}{e^{V_{inD}/nU_t} + 1} = \tanh\left(\frac{V_{inD}}{2nU_t}\right)$$

[3]

Transonductance: $g_m = \frac{I_{tail}}{2nU_t}\sec^2\left(\frac{V_{inD}}{2nU_t}\right) \approx \frac{I_{tail}}{2nU_t}$

Transconductance efficiency: $g_m/I_{tail}=1/2nU_t$ [1]

c) Figure 1.2 shows a cascade of amplifiers where each stage uses the circuit from Figure 1.1. The total noise contribution in each stage is due to the two input devices of the differential pair and the gain of each stage is given by $G_i = -g_m R_{out}$, where g_m is the transconductance of the differential pair in Figure 1.1 and $R_{out} = 100$ GΩ.

i) Given the bias current I_{tail}=20nA, calculate the minimum area A_1 required by the first stage, such that the total contribution of the noise at the output from the first stage is less than 500mV. You may assume $K_w = 1\text{x}10^{-30}$ V^2C, $K_f = 3.5\text{x}10^{-21}$ Hz/F^2, ΔF =100 Hz, f_h = 100Hz, f_l = 1Hz and the transconductance is constant over a given input voltage.

Solutions

$$v_{no}^2 = \sum_{i=1}^{i=M} v_{ni}^2 G_i^2 = \sum_{i=1}^{i=M} \left(n_i \frac{K_w}{\left(I_i / n_i\right)^p} \cdot \Delta f + n_i \frac{K_f}{\left(A_i / n_i\right)} \cdot \ln\left(\frac{f_h}{f_l}\right) \right) G_i^2$$

Solution:
ni=2, P=1
I_{tail}=20nA
$g_m = I_{tail}/2nU_t$ = 20nA/(2x1.23x0.026) = 0.312 uS [1]
Voltage gain $G_i = -g_m R_{out}$ =-0.312uS x 100 GΩ = -31200
Input noise requirement for four stages: $V_{n1}^2 x G_1 x G_1 < (500mV)^2$
$V_{n1}^2 x 973.44x10^6 < (500mV)^2$
$V_{n1}^2 = 0.256821x10^{-9}\ (V)^2$

$$V_{n1}^2 < 0.256821x10^{-9}\ V^2 < \left(n_i \frac{K_w}{\left(I_i / n_i\right)^p} \cdot \Delta f + n_i \frac{K_f}{\left(A_i / n_i\right)} \cdot \ln\left(\frac{f_h}{f_l}\right) \right)$$

$0.256821x10^{-9}\ V^2 < 1.6e\text{-}20 + 2.5e\text{-}19/A$
Solving for A_1 gives $A > 973.4\text{-}12\ (m)^2$ [1]

ii) Propose suitable gate widths and lengths for transistors M1 and M2 for a typical 65nm process.
$A > 973.4\text{-}12\ (m)^2$ so, W=35um, L=35um is a possible solution. [2]

[2]

iii) Given that the system shown in Figure 1.2 operates on a Bluetooth signal at 2.4 GHz, but conveys audio information at 44 kHz, propose one method to reduce the power consumption requirements before quantizing the analogue signal.

Using a mixer you can downcovert the 2.4GHz signal to 44KHz. This will reduce the quantiser requirements from 4.8Gsamples/sec to just 88Ksamples/sec assuming we are sampling at nyquist. [2]

[2]

Solutions

2. This question concerns translinear circuits.

a) Explain why MOSFETs operating in weak-inversion make suitable translinear elements and state two limitations when using these in translinear circuits which may apply specifically to weak inversion operation.

[bookwork]
MOSFETs in weak inversion have an exponential relationship between VGS and IDS. The trancoductance is therefore linear with VGS. Through the logarithmic relationship we can therefore multiply currents in a KVL loop. [2]

Drawbacks:
Currents need to be limited to below 100nA. [1]
Temperatures must be equal to get rid of thermal voltage terms. [1]

[4]

b) Figure 2.1 shows a translinear circuit.

i) Derive the transfer function I_{out} of this circuit using the translinear principle. You may assume that currents I_1 and I_2 are input currents and I_4 is a static bias.

Translinear Loop: $I_{Q1}.I_{Q2}.I_{Q3}.I_{Q4}.I_{Q5}=I_{Q6}.I_{Q7}.I_{Q8}.I_{Q9}.I_{Q10}$ [2]
Also $I_{Q1}=I_{Q2}=I_{out}-I_2$, $I_{Q4}=I_{Q5}=I_{out}$, $I_{Q6}=I_{Q7}=I_2$, $I_{Q10}=I_{Q9}=I_4$
Substituting yields:
$I_{out}=I_4\sqrt{(I_3/I_1)}$ [3]

[5]

ii) Write down an expression for the output current I_{out} when I_{in}= Asin(ωt) and I_1 and I_3 are derived as:

$$I_3 = \left|\frac{dI_{in}}{dt}\right|, \qquad I_1 = \left|\int I_{in}\, dt\right|$$

[4]

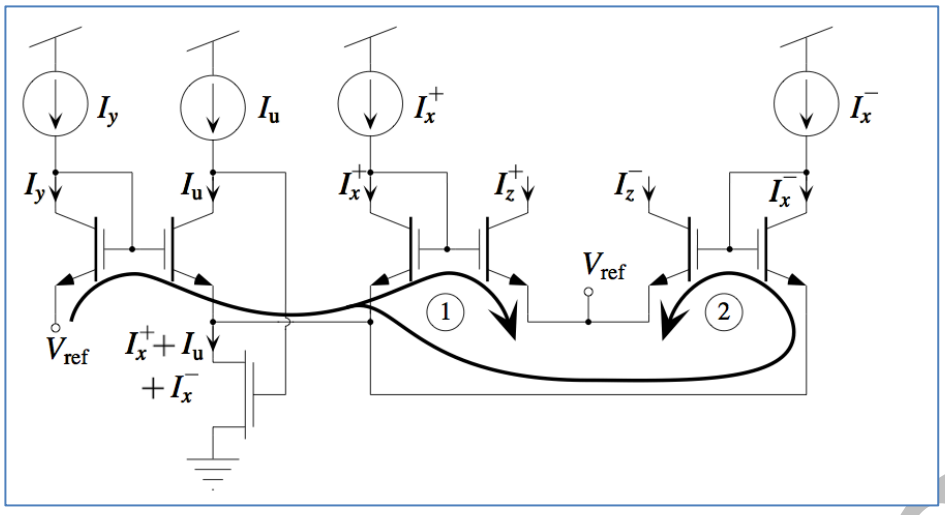
$I_3 = \left|\frac{dI_{in}}{d_t}\right| = A\omega cos(\omega t)$, [1]
$I_1 = |\int I_{in}\, dt| = \frac{A}{\omega}\cos(\omega t)$ [0.5]
$I_{out}=I_4 \times \omega$ [2]

c) Synthesise using Type A translinear cells, a multiplier circuit that implements the following function:

$$z^+ - z^- = (x^+ - x^-)y$$

Draw the complete circuit including all biasing schemes and remove redundant loops.

Solutions

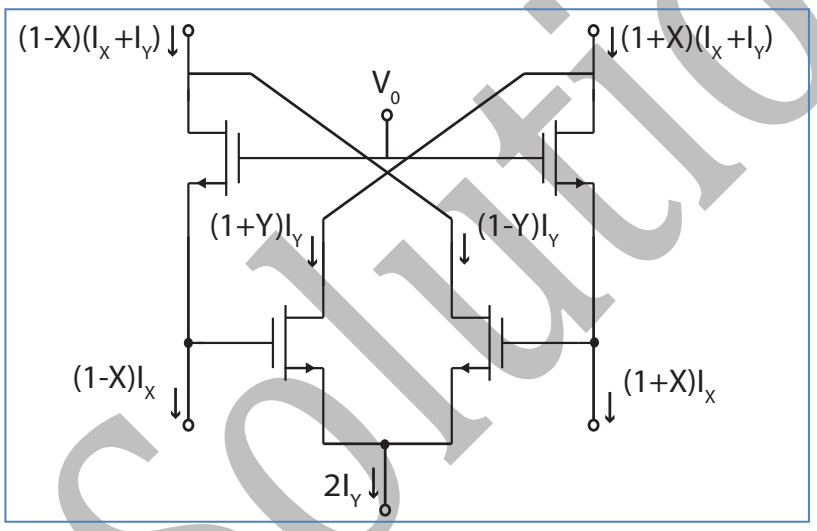


[2] For synthesis equation
[2] For circuit
[1] For removing redundancies.

[5]

d) Explain one advantage of a Gilbert gain cell over a conventional translinear multiplier.

Bookwork:



Advantage: Reuses output current to provide more gain [2]

[2]

Solutions

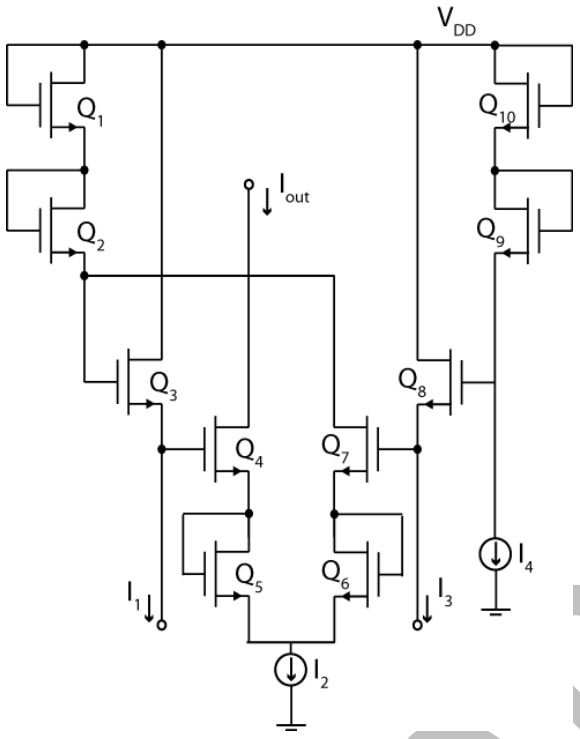
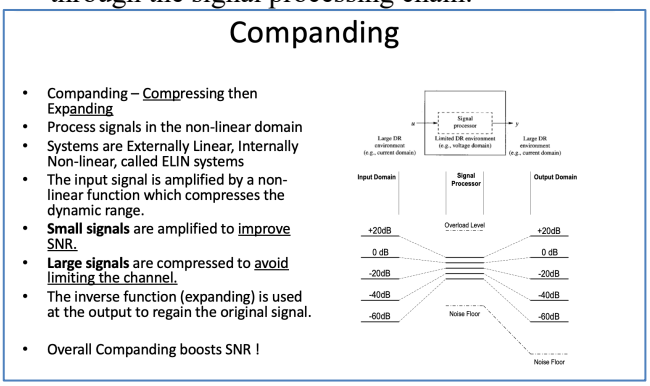


Figure 2.1

Solutions

3. a) Explain the principle of companding, illustrating how the dynamic range changes through the signal processing chain.



[4]

b) Figure 3.1 shows a block diagram of a current-mode companding integrator, with an input X and an output Y. f() and g() represent non-linear functions which compress and expand the signal such that the input-output relationship of the integrator is linear and given by:

$$Y = \tau \int X \cdot dt$$

Given that the expansive function is defined as $I_{out} = g(V_c) = I_0 \sinh(V_c/nU_t)$, derive the function f(x) such that linear integration is achieved. All constants should be grouped to represent a current I_1, with the function f(x) composed of just I_{in}, I_1 and I_{cosh}, whereby $I_{cosh} = I_0 \cosh(V_c/nU_t)$.

New derivation:

$$Y = I_{out} = I_0 \sinh(V_c / nU_t) = g(V_C)$$

$$\frac{d(g(V_c))}{dV_C} = \frac{dY}{dV_C} = \frac{dI_{out}}{dV_C} = \frac{I_0}{nU_t}\cosh(V_C / nU_t) = \frac{I_{Cosh}}{nU_t}$$

$$f(x) = I_C = X\tau C\left(\frac{dY}{dV_C}\right)^{-1} = X\tau C\frac{nU_t}{I_{\cosh}} = X\frac{I_1}{I_{\cosh}} \qquad X = I_{in}, \quad I_1 = \tau \cdot CnU_t$$

$$I_C = f(I_{in}) = \frac{I_{in}I_1}{I_{\cosh}}$$

[6]

c) The transfer function of an oscillator may be defined in state space representation by the following equations:

$$\dot{X}_1 = -\omega_0 X_1 + \omega_0 X_2$$

$$\dot{X}_2 = -2\omega_0 X_1 + \omega_0 X_2 + \omega_0 U$$

$$Y = X_1$$

where Y is the output, U is the input and X_1 and X_2 are the states.

Solutions

(i) By using the mappings below show how these linear equations can be mapped to non-linear log-domain design equations. You may use the following mapping for constant currents, $I_1 = I_2 = I_\omega$ and $I_U = I_O$.

$$X_1 = I_1 \exp\left(\frac{V_1}{nU_t}\right) \qquad X_2 = I_2 \exp\left(\frac{V_2}{nU_t}\right) \qquad U = I_U \exp\left(\frac{V_U}{nU_t}\right)$$

[5]

Appliction of taught methodology.
Should apply state space mapping and derive the following KCL equations:

$$C\dot{V}_1 = -I_\omega - I_\omega \exp\left(\frac{V_2 - V_1}{nU_t}\right)$$

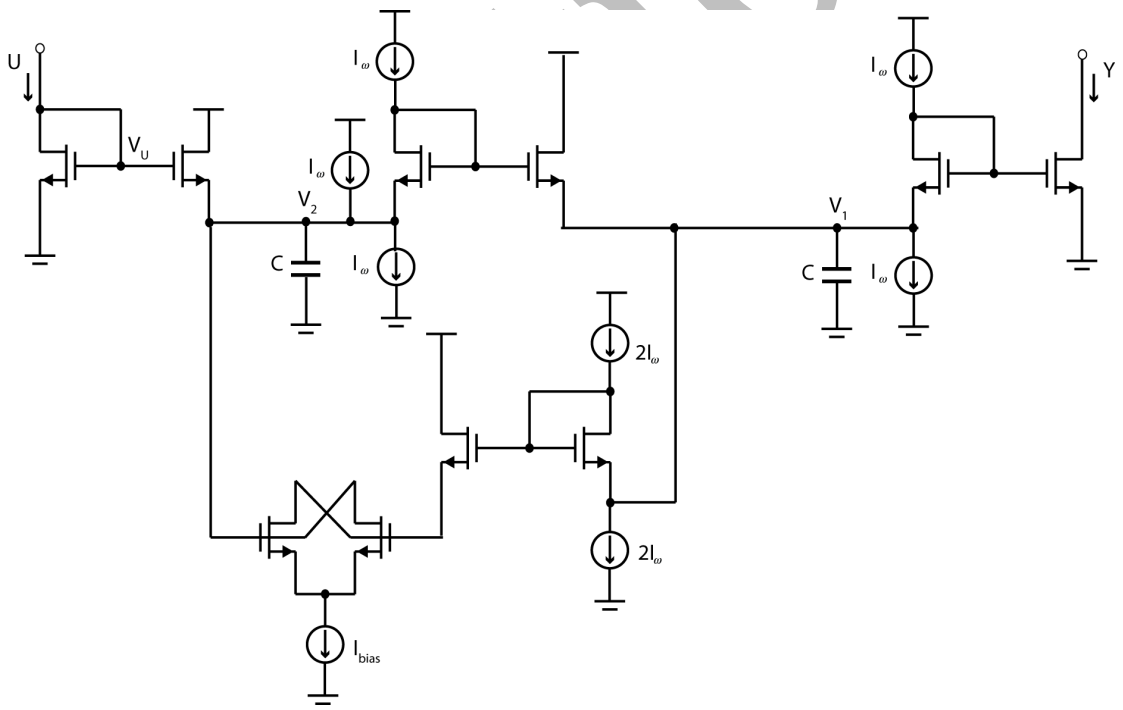
$$C\dot{V}_2 = -2I_\omega \exp\left(\frac{V_1 - V_2}{nU_t}\right) + I_\omega + I_0 \exp\left(\frac{V_U - V_2}{nU_t}\right)$$

$$Y = I_\omega \exp\left(\frac{V_1}{nU_t}\right)$$

[5]

(ii) With these log-domain design equations, sketch a schematic of the final log domain filter using weak inversion MOS transistors.

Drawing of log domain circuit.



[5]

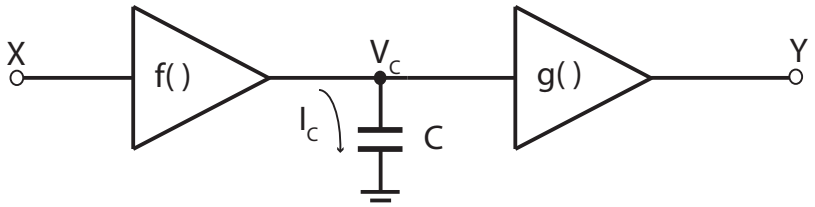


Figure 3.1

Solutions

4.

a) Figure 4.1 shows a conventional second-generation positive current conveyor, CCII+.

(i) Explain it's operation principle and describe how impedances at each of its three ports, X, Y, Z, differ from a standard operational amplifier.

[bookwork]

- Current-voltage characteristics
  $V_X=V_Y$, $I_Z=\pm I_X$, $I_Y=0$
  $Z_Y \rightarrow \infty$, $Z_X \rightarrow 0$, $Z_Z \rightarrow \infty$ [1]
- Voltage-follower between Y-input and X-output
- Current-follower between X-input and Z-output [1]

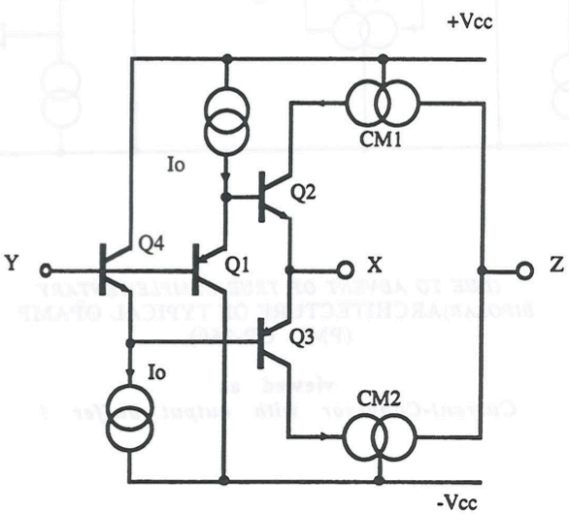
Plus a comparison with Op-amp terminals [1]

[4]

(ii) Draw the circuit of a transistor implementation of a bi-directional CCII+ which does not use any opamps.

Application of taught circuits

Schematic:

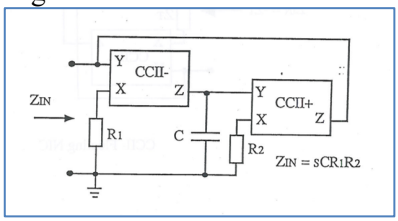


Internal Schematic of CCII01

[4]

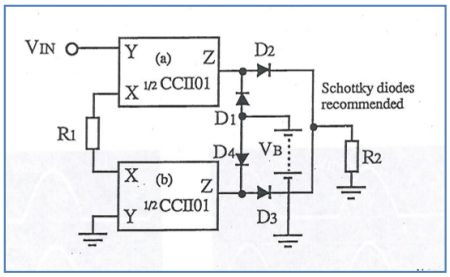
Using current conveyors design circuits which implement:

A grounded inductance.



[2]

A full wave voltage mode rectifier.



[2]

[4]

b) Derive the generalized equation for the output currents, I_{outi} of the circuit in Figure 4.2 and explain the function of this circuit.

Derivation from Theory:

This is a current normalizer. [1]

Derivation [3]:

$$I_{ini} = I_0 \exp\frac{V_{di}}{nU_t}$$

$$I_{outi} = I_0 \exp\frac{V_{di} - V_c}{nU_t} = I_{ini} \exp\frac{-V_C}{nU_t}$$

$$\sum_{i=1}^{N} I_{outi} = I_b$$

$$\exp\frac{-V_C}{nU_t} = \frac{I_b}{\sum_{i=1}^{N} I_{ini}}$$

$$I_{outi} = I_b \frac{I_{ini}}{\sum_{i=1}^{N} I_{ini}}$$

[4]

a. Figure 4.3 shows a circuit which is used a variable OTA. Using your results from part b. or otherwise derive an equation for the output current I_j as a function of the differential input voltage (E_j-V). You may assume all transistors are operating in weak inversion.

Derivation of new Theory:

For differential pair on the right:

$$I_1 - I_2 = I_{at} \tanh\left(\frac{E_j - V}{2nU_t}\right) = I_0 \exp\left(\frac{v_{d1}}{Ut}\right) - I_0 \exp\left(\frac{v_{d2}}{Ut}\right)$$

For differential pair on the left:

$$I_j = I_3 - I_4 = I_0 \exp\left(\frac{v_{d1} - v_c}{nUt}\right) - I_0 \exp\left(\frac{v_{d2} - v_c}{nUt}\right)$$

$$I_j = \left(I_0 \exp\left(\frac{v_{d1}}{nUt}\right) - I_0 \exp\left(\frac{v_{d2}}{nUt}\right)\right) \exp\left(\frac{-v_c}{nUt}\right)$$

$$I_j = \left(I_0 \exp\left(\frac{v_{d1}}{nUt}\right) - I_0 \exp\left(\frac{v_{d2}}{nUt}\right)\right) \frac{I_{st}}{I_{at}}$$

$$I_j = I_{at} \tanh\left(\frac{E_j - V}{2nU_t}\right) \exp\left(\frac{1}{n}\right) \frac{I_{st}}{I_{at}}$$

[4

Solutions

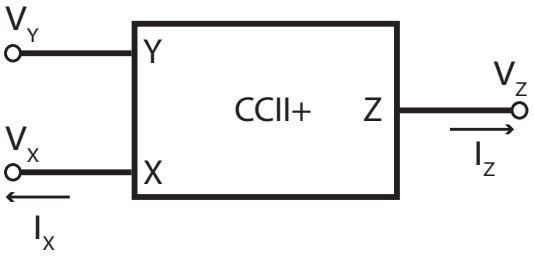


Figure 4.1

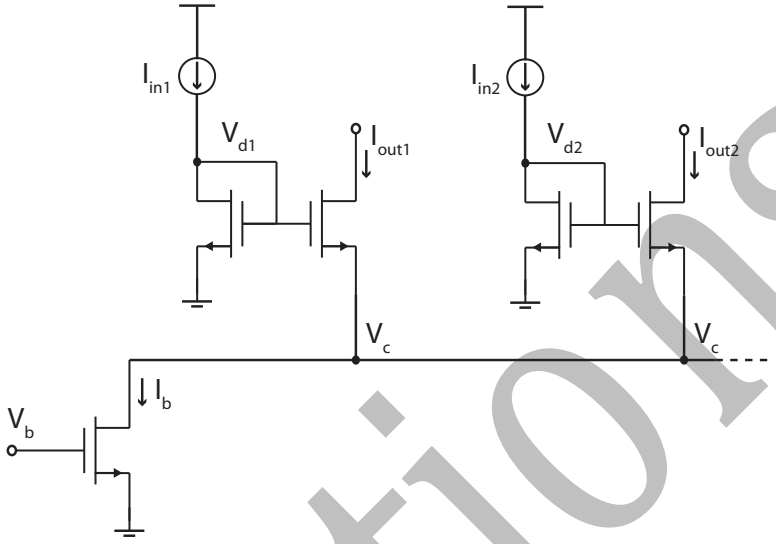


Figure 4.2

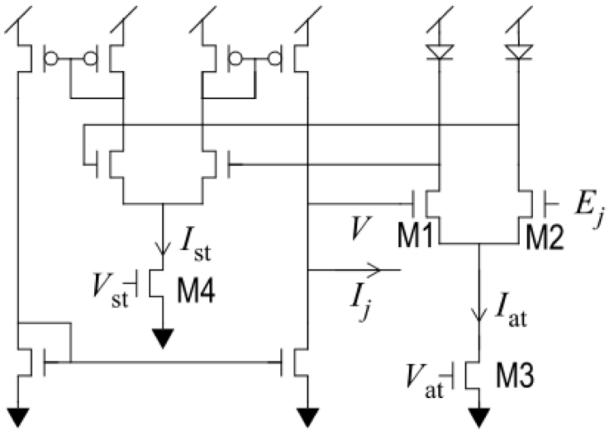


Figure 4.3

Solutions

5.

a) Explain the principle of autozeroing with the aid of diagrams.

[bookwork]

Basic Principle:

- Sample the unwanted quantity (noise and offset)
- Subtract this from the instantaneous value of the contaminated signal.
- The cancellation can occur at the input , or at an intermediate node between the input and output.
- Constant noise (i.e offset) will get subtracted.

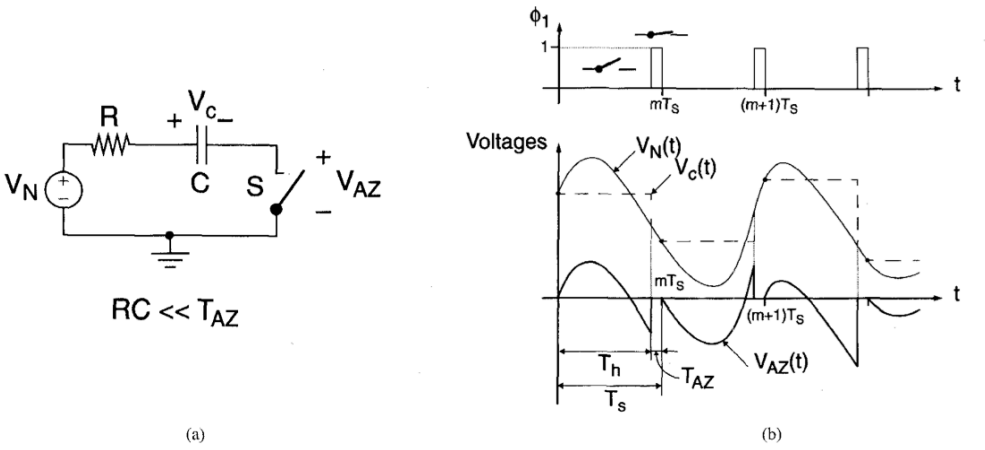


Fig. 2. (a) Basic AZ circuit and autozeroed signal: (b) shows voltages in (a).

[3]

b) Explain how autozeroing affects the noise spectrum of the signal.

Low-frequency random noise (i.e 1/f noise) will be high pass filtered and thus reduced at low frequencies at the expense of increased wide-band noise due to aliasing.

[3]

[2]

c) Figure 5.1 shows a switched capacitor amplifier correlated double sampling amplifier.

(i) Derive the output voltage of this amplifier V_{out} for both phases Φ_1 and Φ_2 showing how this removes any offset in the amplifier. You may ignore feedback capacitor C_{dg}.

Derivation from theory:

$\Phi 1$: C_1 charges to V_{in}-V_{os}, C_2 charges to V_{os}, $V_{out}=V_{os}$

$\Phi 2$: C_1 charges to -V_{os}, C_2 charges to V_{os}-V_{out}

Charge conservation on node A: $Q'_1-Q_1=Q'_2-Q_2$

$C_1(V_{in}-V_{os})-C_1(-V_{os})=C_2V_{os}-C_2(V_{os}-V_{out})$

$V_{out}=V_{in}C_1/C_2$ independent of V_{os} !

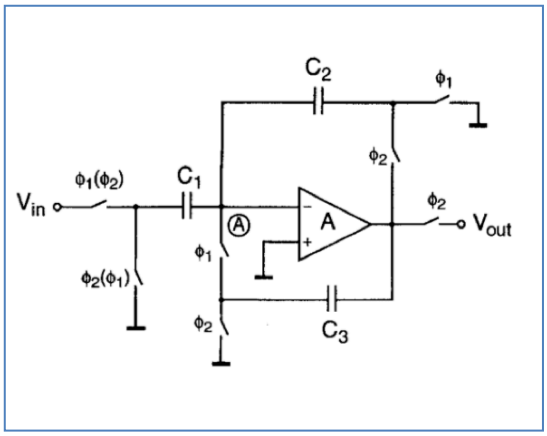
[Total 4 points] [5]

(ii) Show how the circuit of Figure 5.1 can be modified to reduce the high slew rate requirements, and derive the change in output voltage after this modification.

Application of taught Theory:

Circuit with feedback capacitor C3.

Solutio



[3]

Final change in output voltage, $\Delta V=V_{out}(\Phi 1)-V_{out}(\Phi 2)= (Q_3(\Phi 1)-Q_3(\Phi 2))/C_3=V_A=V_{os}-V_{out}/A$ [1]

[Total 4 points]

[5]

d) Derive the output current, I_{out} for the switch current algorithmic cell given in Figure 5.2 showing how this eliminates any error caused due to the switches.

Derivation from Theory:

Operation

Assuming a charge injection δ

- Period 1: $I_{ds1}-J=\delta 1$
- Period 2: $I_{ds2}-J=-\delta 1+\delta 2$
- Period 3: $I_{ds3}-J=I_{in}+\delta 1-\delta 2+\delta 3$
- Period 4: $I_{ds1}-J=I_{in}+\delta 1-I_{ds3}=+\delta 2-\delta 3$
- Period 5: $I_{ds2}-J=\delta 2-\delta 2+\delta 3=\delta 3$
- Period 6: $I_{ds3}=I_{in}+\delta 3-\delta 3=I_{in}$
- Period 7: $I_{out}=I_{in}$!

[5]

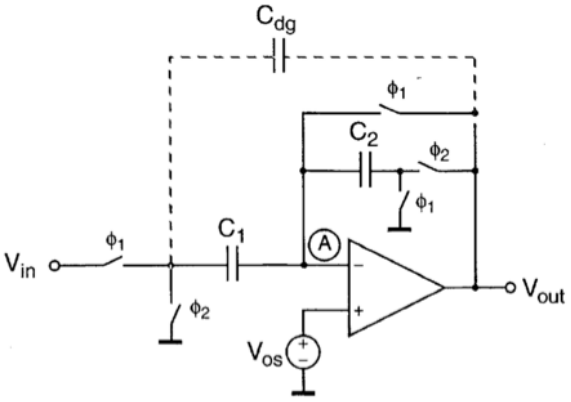


Figure 5.1

Solutions

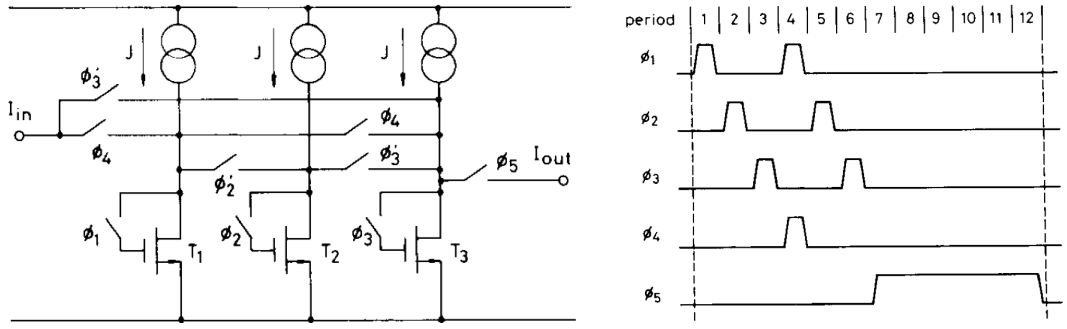
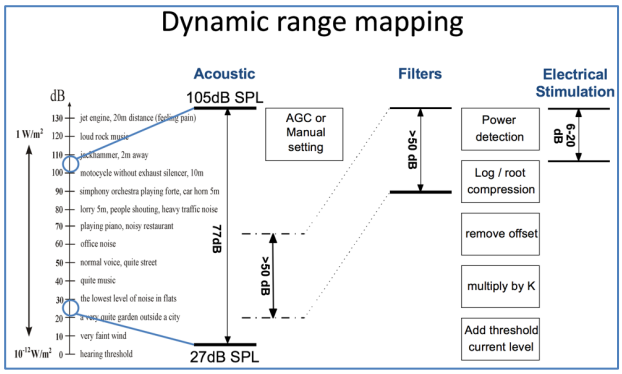


Figure 5.2

Solutions

6. Figure 6.1 shows the block diagram of a cochlear prosthesis.

a. Explain the concept of dynamic range mapping used in this system and why it is beneficial.



- Dynamic range mapping Reduces dynamic range for processing.
- Therefore enables low-power operation for the rest of the analogue processing due to more relaxed dynamic range requirement.
- The instantaneous dynamic range of speech is less than 60db.
- Compress 80dB from AFE to 60 dB in AGC.

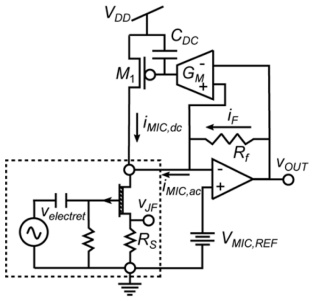
[5]

b. Sketch the schematic of a suitable low-power Analogue Front End (AFE) which is suitable for audio frequencies and explain it's operation.

[5]

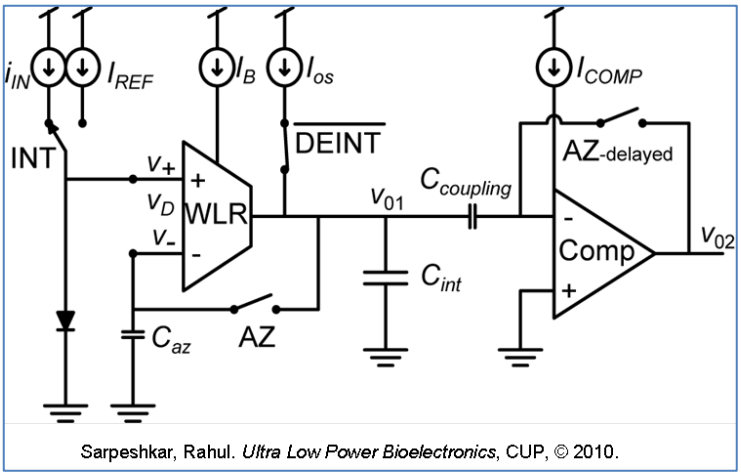
Low-power microphone audio front end

- Sense current of JFET, $I_{MIC,REF}$ using a transimpedance amplifier.
- Increasing R_f will increase gain and SNR.
- JFET has in-built DC bias of 20uA which may saturate the output however.
- Shunt an equivalent DC current, $I_{MIC,DC}$, in input which carries no audio information.
- $G_M C$ negative feedback loop with a cut off <100Hz establishes this and high pass filters the output.
- Also attenuates 1/f noise !
- Only is AC audio information is left as $I_{MIC,AC}$ for amplification.



Sarpeshkar, Rahul. *Ultra Low Power Bioelectronics*, CUP, © 2010.

c. Sketch the schematic of a suitable logarithmic analogue to digital converter which also uses autozeroing to remove any offset in the amplifiers.



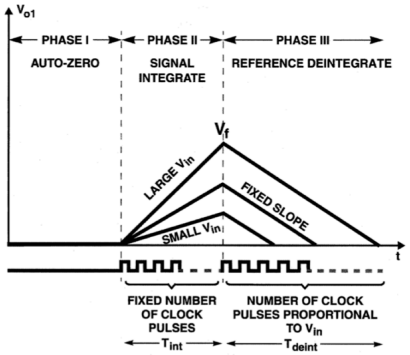
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[5]

Explain it's operation showing timing diagrams and equations where necessary.

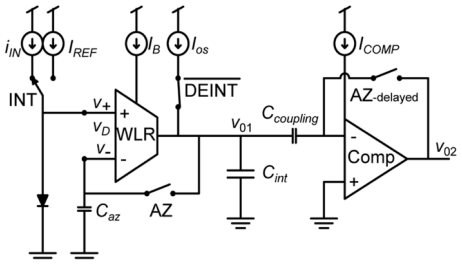
Micropower log-ADC

- Uses a dual slope algorithm.
- Exploits successive integration /de-integration strategy.
- On integration charge capacitor C_{int} with a current proportional to input.
- Then discharge on de-integration phase with a fixed current I_{os}.
- Count time it takes for I_{os} to discharge C_{int}.
- Evaluated by counting clock cycles in a counter.



Operation – Three phases

- **Phase I**: V_+ is tied to I_{REF} and switch AZ is closed, so the offset of the WLR is stored on C_{AZ} and the offset of the comparator is stored on $C_{coupling}$. I_{OS} ensures V_- has a negative offset for de-integration phase.
- **Phase 2:** AZ is opened, and V_+ is switched to I_{IN}, which causes the WLR to charge up C_{int} with a current I_{int} proportional to the log(i_{in}), for a fixed time T_{int}. $I_{int}=g_m(V_D)$ where $V_D=V_+-V_-$
- **Phase 3:** V_+ is switched to I_{REF}(while AZ stays open), and I_{OS} is disconnected, which causes the WLR to discharge C_{int} back down to ground, with a fixed current $I_{deint}=g_m(V_D)$, because V_D will now be negative.
- The time it takes to discharge, T_{deint} is then counted with a digital counter.
- Since T_{deint} is a result of the ratio of two currents, any temperature dependence cancels out !



$$V_f = \frac{I_{\text{int}}}{C_{\text{int}}} \times T_{\text{int}} \quad (I \rightarrow V \text{ transformation})$$

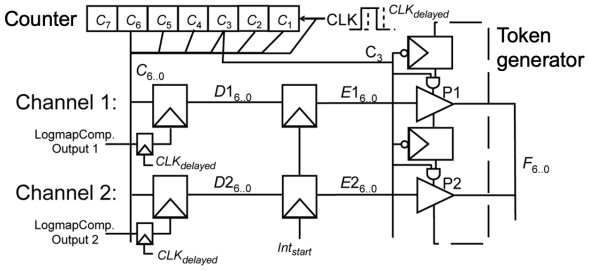
$$T_{\text{deint}} = \frac{V_f}{I_{\text{deint}}/C_{\text{int}}} \quad (V \rightarrow t \text{ transformation})$$

$$= \frac{I_{\text{int}}}{I_{\text{deint}}} \times T_{\text{int}}.$$

d)

CIS scanner

- Continuous Interleaved Sampling (CIS).
- Used to generate currents to stimulate 16 electrodes.
- Converts outputs of log-ADC to bi-phasic pulses for stimulation.



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[Total 5 points]

Solutions